

Li

28052 Linda Ln, Santa Clarita, CA 91350 | 661-476-2715 | liyihong0512@gmail.com

Summary

A self-motivated engineer proficient in JavaScript, Python, CSS, and HTML, with a passion for computer science, electronic music production, and digital art. Leverages independent study to master diverse technical and creative disciplines. Eager to contribute a unique blend of skills and enthusiasm to the team

Programming Languages: C, Python, Java, Javascript, CSS, HTML

Education

College of the Canyons | Santa Clarita, CA

Bachelor of Science in Computer Science and Music

Sep 2024 - May 2028

GPA: 4.00/4.00

Honor: Dean's List

Courses: Data Structures, Architecture/Assembly Language, Intro Algorithms and Prog/Java, C programming, C++ Obj. Oriented Programming

Project Experience

Minecraft Plugin Development | Santa Clarita, CA

Nov 2023 – Dec 2024

- Created a plugin to divide players into groups and enable messaging for both groups and individual players
- Built a repository to list and manage MCDReforged plugins
- Develop a plugin that assembles commands into stacks and automates their execution

Fully Connected Neural Network, FCNN Development | Santa Clarita, CA

Jun 2024 – Jul 2024

- Developed a fully connected neural network, trained with the MNIST dataset, to convert handwritten digits in images into digital numbers on a computer

Dynamic Memory Allocation Program Development | Santa Clarita, CA

Sep 2023 – Jun 2024

- Developed a dynamic memory allocation program using assembly language, implementing the Best Fit algorithm to efficiently manage memory allocation and deallocation

Browser Plugin Development | Santa Clarita, CA

Nov 2023 – Jun 2024

- Built a browser extension to help organizing tabs using JavaScript, CSS, and HTML
- Transformed the traditional flat tab display by layering tabs, increasing information density per tab, and improving user efficiency in locating tabs
- Conducted functional testing with peers, identifying and resolving bugs to ensure plugin's robustness
- Expanding features with plans to launch a public beta on Google Chrome and Microsoft Edge browser

Web Application Development | Santa Clarita, CA

Oct 2023 – Dec 2023

- Build a Cam-and-Petra-Mcgee- Web app for email management featuring ranking, auto-responses, an email bot, and algorithm integration, along with video trend analysis

Intelligent Driving - Mobile Robot Design and Development

Jul 2022 – Oct 2022

- Designed the structure of a robot that could climb the stairs and deliver small items using Fusion 360
- Optimized drive ratios, weight distribution, and mechanical design for improved joint efficiency, stability, and load resistance

Research Experience

Self-learning and promoting

Oct 2022 – Present

- Self-studied "The Elements of Computing Systems" (Nand to Tetris), gaining a comprehensive understanding of computer architecture and the fundamental principles of digital systems
- Self-studied an MIT Press book "Deep Learning" for neural networks, gaining an understanding of the theory, practical applications, and advanced research methods in areas such as mathematics, computer vision, and natural language processing, entered the field of machine learning in general and deep learning in particular
- Self-studied and mastered FL Studio 20 to create electronic music, demonstrating a strong aptitude for learning and applying new software tools

Leadership Experience

Minecraft Server Owner | Personal Project

Jan 2022 – Present

- Led a team of players to construct and manage the Minecraft server environment
- Moderated meetings to discuss and implement community-driven projects, ensuring smooth collaboration
- Managed server performance, resolved technical issues, and implemented updates to optimize user experience
- Organized core server members to plan and develop custom mods and plugins to enhance server functionality